
Engagement-Aware Movie Recommendation

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Abstract

Streaming platforms increasingly rely on recommendation systems to drive user engagement, yet most existing approaches optimize primarily for accuracy while overlooking diversity and novelty. In this work, we study sequential movie recommendation using a real-world Netflix UK clickstream dataset comprising 649,250 interactions from 161,618 users across 8,411 movies. We evaluate three model families: transformer-based sequential models, graph-based collaborative filtering (LightGCN), and a hybrid sequential-graph architecture. BERT4Rec achieves the best overall performance, with bidirectional attention proving particularly effective for short user interaction histories. Additionally, GRU-LightGCN outperforms standalone LightGCN, demonstrating that combining sequential and graph-based signals leads to richer user representations. Beyond accuracy, we measure intra-list diversity and find that stronger models also produce more diverse recommendations, suggesting that improved representation learning can mitigate popularity bias. These results underscore the importance of aligning model architecture with the structural characteristics of user interaction data.

1. Introduction

Recommender systems are a critical component in driving user engagement, retention, and content discovery for modern streaming platforms. In particular, sequential recommendation and next-item prediction based on user historical behavior have gained traction due to being inherently dynamic modeling user preferences over time. Traditional approaches to this problem have focused primarily on modeling user-item interaction sequences using collaborative filtering signals. However, such methods often struggle in realistic settings characterized by sparse data, short interaction histories, and limited contextual understanding of items.

In this project, we investigate engagement-aware sequential movie recommendation and improving both the relevance as well as long-term engagement of recommendations. Specifi-

cally, we explore:

1. Can integrating content-based features, such as movie metadata and user reviews, improve sequential recommendation performance beyond interaction-only models?
2. How can user engagement patterns be incorporated into representation learning to better capture evolving preferences?
3. Do hybrid architectures that combine sequential modeling with graph-based collaborative filtering provide measurable gains over standalone approaches?

To address these research questions, we systematically compare various recommender architectures on a sparse real-world clickstream dataset. The models are evaluated on their output of top- $k \in \{5, 10, 20\}$ recommendations to determine whether incorporating content and engagement features improves both ranking performance and recommendation quality.

1.1. Related Work

1.1.1. SEQUENTIAL RECOMMENDATION FOUNDATIONS AND RELEVANT MODELS

Early work in sequential recommendation focused on modeling user behavior through Factorizing Personalized Markov Chains to capture both user preferences and item-to-item transitions (Rendle et al., 2010). The introduction of deep learning led to RNN-based models which model session-based interactions using gated recurrent units. GRU4Rec demonstrated strong performance in capturing temporal dynamics, influencing our decision to treat recommendation as a sequence prediction problem (Hidasi et al., 2015). However, earlier implementations have their limitations in handling long-range dependencies. Parallelization motivated the shift toward attention-based models in our work.

Transformer architectures have become the dominant paradigm for sequential recommendation due to their ability to model long-range dependencies via self-attention (Vaswani et al., 2017). SASRec applied unidirectional self-attention to capture user preferences from past interactions, demonstrating strong performance especially in sparse set-

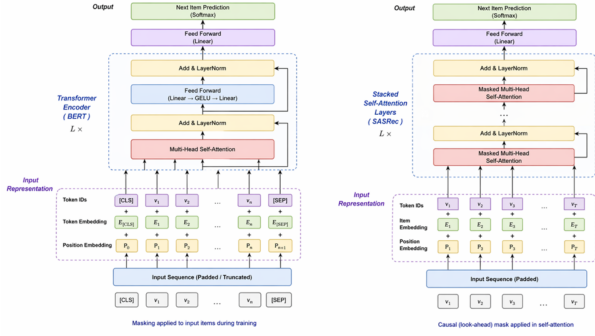


Figure 1. Architecture comparison of BERT4Rec (left) and SAS-Rec (right).

tings. This model directly influenced our use of attention-based sequence modeling as a core component. BERT4Rec introduced bidirectional self-attention via a masked item prediction objective, allowing the model to leverage both past and future context within sequences. This approach significantly improved representation learning and inspired our adoption of bidirectional contextual modeling and deeper architectures for sequence understanding. Our work extends this direction by augmenting transformer models with side information (metadata, engagement features) and modifying training objectives to better align with recommendation quality metrics beyond accuracy.

Graph-based methods provide a complementary perspective by modeling user–item interactions as a bipartite graph. LightGCN simplifies graph convolutional networks to focus on embedding propagation, capturing higher-order collaborative signals (He et al., 2020b). This model influenced our inclusion of graph-enhanced embeddings to encode long-range relationships between users and items. However, graph-based methods lack explicit modeling of temporal order. This limitation motivated our use of hybrid architectures, where graph-derived embeddings are combined with sequential models to capture both collaborative structure and temporal dynamics. Recent work explores hybrid models that combines such paradigms in attempt to bridge the gap between temporal dynamics and collaborative filtering but often optimize primarily for accuracy-based metrics.

1.1.2. CONTENT-AWARE AND REPRESENTATION-ENRICHED MODELS

A key limitation of traditional recommender systems is their reliance on user–item interaction data alone, which leads to degraded performance in sparse or cold-start settings. To address this, recent work has explored incorporating auxiliary content such as item descriptions, metadata, and user reviews to enrich item representations. For instance, BertMoSRec (Xin et al., 2026) integrates BERT-based em-

beddings derived from item descriptions, labels, and reviews with sequential recommendation models, demonstrating improved performance by capturing semantic relationships between items and user preferences. Similarly, review-aware recommender systems leverage textual feedback to better model user intent and item characteristics, providing richer contextual signals than interaction data alone. These approaches underscore the impact of incorporating semantic and contextual information into recommendation pipelines. In our work, this directly motivates the inclusion of movie metadata (genres, release year, popularity) and user behavioral features (engagement statistics, genre preferences). We aim to improve generalization in sparse regimes and better capture the underlying factors driving user engagement by augmenting standard item embeddings with content-aware features.

1.1.3. SELF-SUPERVISED AND CONTRASTIVE LEARNING APPROACHES

Self-supervised learning has emerged as a powerful paradigm for improving representation learning in recommendation systems without requiring additional labeled data. CL4SRec (Xie et al., 2022) introduces contrastive learning for sequential recommendation by generating augmented views of user interaction sequences and maximizing agreement between them, leading to more robust and generalizable user representations. Similarly, momentum contrastive learning (MoCo) has been adapted from computer vision to recommendation tasks, enabling the use of large negative sample queues to learn richer embedding spaces and capture deeper user preference signals (He et al., 2020a). While these methods improve representation quality, they often operate solely on interaction sequences and do not incorporate auxiliary content features. Nonetheless, they demonstrate the effectiveness of optimizing objectives beyond standard next-item prediction. This insight motivates our exploration of multi-objective optimization and post-processing strategies, including reranking and auxiliary loss design, to balance accuracy with diversity and novelty. Rather than directly adopting contrastive pretraining, we incorporate its underlying principle of learning richer representations through additional supervisory signals into our model design and evaluation framework.

1.1.4. HYBRID MODELS

Hybrid recommendation models aim to combine the strengths of sequential modeling and graph-based collaborative filtering. Previous research efforts have explored the combination of paradigms by integrating graph-derived embeddings into sequential encoders, enabling the model to leverage both local temporal patterns and long-range relational signals such as the User Multi-Behavior Graph Network (UMBGN) (Jia et al., 2023). This work directly

motivates our GRU–LightGCN hybrid architecture. In our setting, user interaction sequences are short and sparse, limiting the effectiveness of purely sequential models, while graph-based models lack ordering information necessary for next-item prediction. By combining GRU-based sequence encoding with LightGCN-derived embeddings through a learned gating mechanism, our model adaptively balances sequential and collaborative signals. This design allows the model to rely more on graph structure for users with limited history and more on sequential patterns for users with richer interaction data, providing a unified framework that addresses the limitations of their respective models.

2. Methods

2.1. Dataset

We use the Netflix UK Movies clickstream dataset (CM & Follows, 2021), consisting of 649,250 interactions from 161,618 users and 8,411 movies. Each interaction contains a user ID, item ID, and timestamp, representing implicit feedback through viewing behavior.

We filter out extreme outliers (users with >37 interactions) and restrict evaluation to users with at least three interactions, yielding 72,138 valid users. Data is split using a leave-one-out protocol: the final interaction is used for testing, the penultimate for validation, and the remaining interactions for training. Evaluation is performed via full ranking over all items.

Property	Value
Total interactions	649,250
Total users	161,618
Total movies	8,411
Valid users (≥ 3 interactions)	72,138
Median sequence length (all users)	2.0
Median sequence length (valid users)	5.0
Item feature dimensionality	78
User feature dimensionality	38

Table 1. Dataset statistics after preprocessing.

2.2. Features

The base clickstream provides only user identifiers, movie identifiers, and timestamps. To enrich model inputs, we construct two supplementary feature tables by querying the TMDB API (tmd, 2024) and aggregating interaction-level statistics.

Item features capture movie-level content and popularity signals. Each movie is represented by 28 binary genre indicators (Action, Comedy, Drama, etc.), release year and decade bucket, TMDB runtime, language, vote average, and popularity score, alongside interaction-derived statistics including log click count, unique user count, and repeat interaction ratio giving a 78-dimensional item feature vector

after one-hot encoding of language.

User features capture behavioral patterns aggregated from each user’s interaction history. These include total click count, number of unique items watched, session count and average session length (sessions defined by a 30-minute inactivity gap), preferred viewing hour bucket, weekend viewing ratio, engagement tier (quartile-bucketed by total interactions), and a 28-dimensional genre preference vector computed as the mean of genre indicators across all clicked items, reflecting the user’s accumulated content taste. Both feature tables are constructed using the same user- and item-integer indices across all model training pipelines, ensuring no data leakage between splits.

Based on Table 1, the median interaction sequence length is 2.0 across all users and 5.0 among the 72,138 users with at least three interactions, making this a sparse, short-sequence dataset. This sparsity motivates the use of content and behavioral side features to supplement the limited collaborative signal available from interaction history alone, and distinguishes this setting from benchmark datasets such as MovieLens-1M (Harper & Konstan, 2015) where median sequence lengths are substantially longer.

2.3. Models

We evaluate three model families: sequential, graph-based, and hybrid approaches.

2.3.1. SEQUENTIAL MODELS

We implement SASRec and BERT4Rec as shown in 1, which model user interactions as ordered sequences.

SASRec applies causal self-attention to capture temporal dependencies, while BERT4Rec uses a masked prediction objective with bidirectional attention, enabling richer contextual representations (Sun et al., 2019). Given the short sequence lengths in our dataset, BERT4Rec is expected to extract more signal per interaction.

We extend both models by incorporating item and user features through gated projections. We also apply a popularity-debiased cross-entropy loss:

$$w_i = \frac{1}{\sqrt{f_i}}$$

to reduce bias toward frequently interacted items. Models are trained with Pre-LayerNorm transformers and a deeper prediction head for improved stability and capacity.

2.3.2. GRAPH-BASED MODELS

We implement LightGCN (He et al., 2020b) over a bipartite user–item interaction graph $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$, where an edge $(u, i) \in \mathcal{E}$ is added for each observed training interaction. LightGCN removes non-linear activations and learnable

weight matrices at each layer, retaining only symmetrically normalised neighbourhood aggregation — a design choice shown empirically to match or outperform heavier GCN variants in collaborative filtering (He et al., 2020b):

$$e_u^{(l)} = \sum_{i \in \mathcal{N}(u)} \frac{1}{\sqrt{|\mathcal{N}(u)| |\mathcal{N}(i)|}} e_i^{(l-1)}, \quad (1)$$

$$e_i^{(l)} = \sum_{u \in \mathcal{N}(i)} \frac{1}{\sqrt{|\mathcal{N}(i)| |\mathcal{N}(u)|}} e_u^{(l-1)} \quad (2)$$

Final node embeddings are mean-pooled across all L layers: $e_u = \frac{1}{L+1} \sum_{l=0}^L e_u^{(l)}$, $e_i = \frac{1}{L+1} \sum_{l=0}^L e_i^{(l)}$. Rather than predicting directly, LightGCN acts as a feature extractor: its graph-enriched item embeddings are precomputed once per epoch and injected into sequential models via a learned gate $\tilde{e}_i = g \odot e_i^{\text{graph}} + (1-g) \odot e_i^{\text{base}}$, where $g = \sigma(W_g[e_i^{\text{graph}}; e_i^{\text{base}}])$. This decoupling avoids backpropagating through sparse graph operations each step, improving efficiency and stability. LightGCN is especially valuable here given our median sequence length of five: graph propagation surfaces relevant items for sparse users by leveraging population-level co-interaction patterns that purely sequential models cannot see.

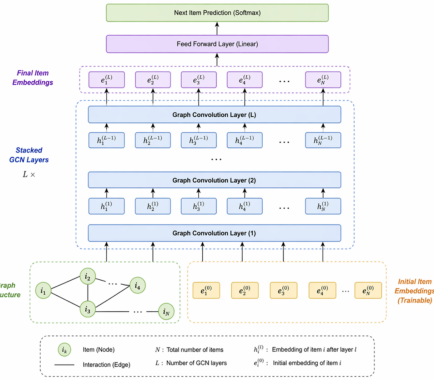


Figure 2. LightGCN architecture. Trainable user and item embeddings are propagated over K layers of normalised neighbourhood aggregation on the bipartite interaction graph. Final embeddings e^* are mean-pooled across layers and injected into downstream sequential models via a learned gate.

2.4. Hybrid Models

Our hybrid architecture combines a two-layer GRU sequential encoder with LightGCN graph-enriched item embeddings to jointly exploit temporal ordering and global collaborative structure, addressing the limitations of each model class in isolation. Each item v_t in the interaction sequence is represented as the sum of a trainable identity embedding,

a positional embedding, and an optional item feature embedding, giving input $x_t = E_{\text{item}}(v_t) + E_{\text{pos}}(t) + E_{\text{feat}}(v_t)$. This sequence is processed by a GRU (Cho et al., 2014) whose final hidden state $h_t = \text{GRU}(x_t, h_{t-1})$ serves as a compact summary of the user’s sequential intent. In parallel, LightGCN propagates embeddings over the user–item bipartite graph following the formulation in Section ??, producing graph-enriched item embeddings $g_t \in \mathbb{R}^d$ precomputed once per epoch to avoid backpropagating through sparse graph operations. The architecture is illustrated in Figure 3.

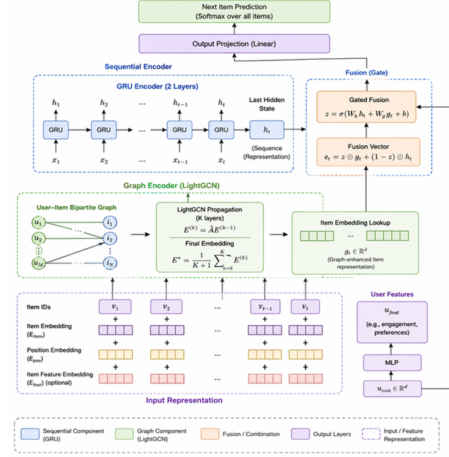


Figure 3. Architecture of the GRU–LightGCN hybrid model. Item identity, positional, and feature embeddings are summed and processed by a two-layer GRU. In parallel, LightGCN produces graph-enriched embeddings over the user–item bipartite graph. The two representations are fused via a learned gate to produce the final prediction via softmax.

The sequential and graph representations are fused via a learned gating mechanism that adaptively weights each encoder’s contribution:

$$z = \sigma(W_h h_t + W_g g_t + b), \quad (3)$$

$$e_t = z \odot g_t + (1 - z) \odot h_t \quad (4)$$

where $W_h, W_g \in \mathbb{R}^{d \times d}$ are learnable projection matrices and $\odot$ denotes element-wise multiplication. When z is close to one the model emphasises graph-collaborative signals; when close to zero it relies on sequential patterns — allowing the model to fall back on population-level structure for users with very short histories. The fused vector e_t is passed through a linear output projection and softmax over all items for next-item prediction, trained with the same popularity-debiased cross-entropy loss and AdamW cosine annealing schedule used across all models (Loshchilov & Hutter, 2019; 2017).

2.5. Training

All models are trained using AdamW with a cosine annealing schedule and warm restarts. Hyperparameters (embedding size, number of layers, learning rate) are selected via grid search on the validation set.

2.6. Evaluation

We report recommendation quality using Hit Rate (HR@K), Normalized Discounted Cumulative Gain (NDCG@K), and Mean Reciprocal Rank (MRR) at $K \in \{5, 10, 20\}$. We also measure diversity which is estimated as the mean pairwise cosine similarity among the top-10 recommended items (i.e. lower is more diverse).

3. Results

3.1. Training Dynamics

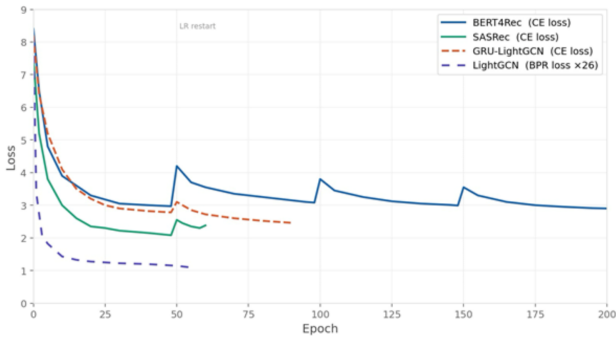


Figure 4. Training loss curves over 200 epochs for all models.

Figure 4 shows training loss over 200 epochs. All models converge stably, with periodic spikes at epochs 50, 100, and 150 corresponding to cosine annealing restarts. These restarts briefly increase loss before recovery, suggesting improved exploration of the loss landscape without destabilizing training. Among sequential models, SASRec converges fastest and to the lowest loss (≈ 2.3), while BERT4Rec converges more slowly and plateaus higher (≈ 2.9), consistent with its more difficult masked prediction objective. The GRU-LightGCN hybrid reaches an intermediate loss (≈ 2.5), reflecting the added complexity of combining sequence and graph representations.

3.2. Quantitative Results

Table 2 reports HR@K, NDCG@K, and MRR across all models. Overall, BERT4Rec achieves the strongest performance across all metrics, followed by SASRec, while graph-based models lag in accuracy but offer greater diversity.

3.3. Discussion

Sequential models outperform graph-based methods. BERT4Rec achieves HR@10 of 0.3085 and NDCG@10 of 0.2081, outperforming SASRec (0.2849 / 0.1853). This confirms that bidirectional attention provides stronger contextual representations, particularly in our short-sequence setting (median length 5). SASRec remains competitive, but its causal masking limits the amount of context available for prediction.

Hybrid models improve over graph-only approaches. LightGCN performs worse than sequential models (HR@10: 0.1636), but the GRU-LightGCN hybrid improves performance (0.1836), indicating that incorporating temporal order provides additional signal beyond collaborative structure alone. However, the hybrid still falls short of transformer-based models, suggesting that sequence modeling remains the dominant factor in this dataset.

Accuracy-diversity trade-off. Graph-based methods produce more diverse recommendations (higher cosine similarity reduction), while sequential models achieve higher accuracy. This reflects a common trade-off: models optimized for next-item prediction tend to favor popular items, whereas graph propagation introduces broader item coverage. Our popularity-debiased loss partially mitigates this effect but does not eliminate it.

3.4. Error Analysis

We observe two primary failure modes:

(1) Short interaction histories. Users with one or two prior interactions provide insufficient signal for personalization, causing models to default to popularity-based recommendations.

(2) Long-tail items. Items with very few interactions are rarely recommended across all models, even with debiasing. This highlights the limitations of collaborative signals in sparse regimes and suggests the need for richer content representations.

Overall, these results demonstrate that transformer-based sequential models are best suited for sparse, short-sequence recommendation, while hybrid and graph-based approaches offer complementary benefits in diversity and coverage.

4. Process

Our development process involved several exploratory directions beyond the final models, including approaches that failed but informed our design decisions.

Table 2. Test set evaluation results across all models.

Model	HR@5	NDCG@5	HR@10	NDCG@10	HR@20	NDCG@20	MRR
BERT4Rec	0.2626	0.2003	0.3085	0.2081	0.3903	0.2370	0.1966
SASRec	0.2212	0.1647	0.2849	0.1853	0.3497	0.2016	0.1634
GRU-LightGCN	0.1303	0.0910	0.1836	0.1082	0.2477	0.1243	0.0947
LightGCN	0.1103	0.0870	0.1636	0.0982	0.2177	0.1043	0.0847

4.1. Reinforcement Learning (PPO)

We initially explored Proximal Policy Optimization (PPO) applied to a pretrained GRU-LightGCN model, using a multi-objective reward combining accuracy, diversity, and novelty. However, performance degraded drastically (HR@10 dropped from 0.1836 to 0.0029).

This failure stemmed from three factors: (1) low-variance reward signals in a large action space (8,411 items), leading to noisy updates; (2) an untrained critic that failed to learn meaningful value estimates; and (3) instability from repeated policy updates, which corrupted pretrained representations. More fundamentally, next-item recommendation is a single-step prediction task, making RL ill-suited due to the absence of temporal credit assignment.

4.2. Post-Processing Approaches

We then shifted to post-processing methods built on top of pretrained models.

Reranker. A lightweight MLP reranker rescored the top-50 candidates from GRU-LightGCN using item and user features. This achieved the best post-processing performance (HR@10: 0.2173), primarily improving ranking position rather than recall. However, performance was bounded by candidate coverage, with only 38% of true items appearing in the top-50 set.

Auxiliary loss fine-tuning. We introduced diversity, novelty, and KL regularization terms to the training objective. This improved validation performance but showed mild overfitting and limited gains on the test set (HR@10: 0.2132). The novelty term provided negligible benefit, suggesting that popularity bias had already been partially addressed by the base loss.

Direct Preference Optimization (DPO). We reframed ranking as a preference learning problem using hard negatives. While preference accuracy improved, ranking performance deteriorated after early epochs due to overly aggressive probability updates, which destabilized the learned ranking distribution.

4.3. Training Challenges

An early stopping issue in GRU-LightGCN caused training to terminate prematurely due to temporary loss spikes from cosine annealing restarts. As a result, the model was undertrained, limiting its performance and downstream improvements from reranking.

5. Contributions

We used PyTorch for all model implementations and PyTorch Sparse for efficient graph operations. No starter code was used; all implementations were developed independently based on the original papers. We developed our own feature engineering pipelines to construct movie metadata and user behavioral features from the raw click-stream data. Within the team, Illy focused on replicating and adapting the BERT4Rec model to our dataset. Arnav contributed to the implementation of LightGCN and was responsible for extracting and integrating graph-based recommendation embeddings. Sanjay and Rhea led feature construction, hybrid model design, training pipeline development, evaluation, and post-processing methods. All team members contributed equally to composition and refinement of this report. All code used to generate the results in this paper is publicly available at <https://github.com/illydh/csci567-group-project>. The raw dataset can be downloaded from <https://www.kaggle.com/datasets/vodclickstream/netflix-audience-behaviour-uk-movies>.

To reproduce our results, the main steps are:

6. AI Usage

We used Claude (Anthropic) and GitHub Copilot extensively across all stages of the project. For **literature review**, Claude assisted in identifying relevant prior work as well as in understanding architectural differences between models. For **coding**, Copilot accelerated boilerplate implementation (training loops, evaluation harnesses, data loaders) and Claude helped debug non-obvious issues such as the early stopping interaction with cosine annealing restarts and the PPO critic initialization problem. For **writing**, Claude assisted in drafting and refining the Methods and Results

sections of this report. For **ideation**, Claude helped reason through the failure modes of the Actor-Critic RL approach and suggested the reranker as a lower-risk alternative. All experimental results, numbers, and conclusions are our own and were verified directly against our code outputs. We take full responsibility for all claims in this report.

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